

Kuapa Dwaso: Product Vision

Kuapa Dwaso (“Farmer’s Market” in Akan) is a warehouse-based platform for produce aggregation, inventory, sales, and dispatch. Built around the realities of agricultural trade in regions like Tarkwa, it replaces risky, speculative farm-to-city transport with a secure, demand-driven local warehouse model.

Context and System Design

The Problem: Consultations with smallholder farmers and market traders around Tarkwa revealed that the biggest bottleneck in agricultural trade is a fragmented and risky produce movement system. Farmers face high transport costs, poor roads, multiple layers of intermediaries, pressure to sell on credit, and heavy spoilage losses.

Because farming land around Tarkwa is damaged by illegal mining, produce comes from outlying communities and passes through second-chain buyers, loading boys, tricycles, and trucks. Every transfer adds cost and spoilage risk. Worse, a farmer who arrives in the city with perishable goods and no buyer has no bargaining power and often sells on forced credit.

The Solution: Kuapa Dwaso establishes local community warehouses as physical trust points. Farmers deposit produce nearby instead of transporting it to the city on speculation. The platform matches warehouse stock to confirmed city demand and coordinates bulk transport only when orders exist.

Key Design Decisions from Field Insights

Consignment Model	Farmers retain ownership of stored produce and pay a transparent daily storage fee deducted from sale proceeds, with digital tracking of expected payouts. This avoids the capital burden of the platform buying stock upfront.
Demand-Driven Dispatch	Traders submit requirements ahead of market windows (especially Friday mornings), so produce leaves the warehouse only against confirmed orders.
Landed-Cost Transparency	The platform tracks every cost component (produce, loading, tricycle, truck, handling, spoilage buffers) so users understand real margins.
Custody Tracking	The system maps the actual supply chain, including second-chain buyers, aggregators, and local transporters.
Realistic Accessibility	Most farmers have household access to a smartphone, so we built a lightweight web app with agent-assisted registration instead of fragile two-way SMS workflows. One-way SMS handles transactional updates (Produce Received, Produce Sold, Payment Recorded).

Core Product Capabilities

1 Warehouse Intake & Agent Dashboard

Agents register farmers, record intake details (crop, variety, quantity, grade, storage rate, shelf life), attach photo evidence, and generate Storage Receipts and Inventory Batches.

2 Mobile-First Farmer Portal

Farmers see their produce and receipts by warehouse, a transparent storage fee ledger, and clear sale and payout tracking (gross sales, deducted fees, net amount due).

3 Aggregate Buyer Console

Buyers browse combined warehouse stock by crop, grade, location, and dispatch date, place and pay for orders digitally, schedule dispatch around market days, and track order status end to end.

4 Logistics & Dispatch Manager

Consolidates multiple orders into single truck dispatches, tracking vehicles, transit costs, payers, and delivery confirmations.

5 Administration Control Center

Admins manage accounts and warehouses, configure fee rules, review audit logs for sensitive actions, and resolve disputes.

System Workflow & Produce Lifecycle

Stage	Action	Fulfillment Context
Step 1	Intake & Deposit	Farmer delivers produce to Warehouse Agent.
Step 2	Registration	Agent records weight, grade, photo, and storage rate in system.
Step 3	Confirmation	System sends SMS confirmation & Storage Receipt to Farmer.
Step 4	Ordering	Buyer browses available warehouse stock and places Order.
Step 5	Reservation	System reserves inventory for the Order.
Step 6	Payment	Buyer pays digitally via Paystack.
Step 7	Preparation	Agent prepares inventory and links it to a Dispatch.
Step 8	Logistics	System assigns Dispatch task to Transporter.
Step 9	Delivery	Transporter updates transit status (Departed -> Delivered).
Step 10	Deductions	System deducts storage fees & calculates Net Payout for Farmer.
Step 11	Payout	System sends SMS (Payment Ready) to Farmer.

Technical Architecture & Core Stack

To achieve a lightweight experience for farmers, real-time inventory management for agents, and enterprise stability for administrators, Kuapa Dwaso is built with a decoupled monorepo architecture:

Turborepo + pnpm	Keeps multiple client applications and shared packages (types, permissions, configs) in a single, strictly typed codebase for safe, unified updates.
Convex Database	Serves as the system of record. Automatic real-time queries and mutations prevent double-selling of stock across agent and buyer interfaces.
NestJS + Fastify	Powers backend services, processing file uploads, security validation, and managing third-party system webhooks.
Firebase Auth	Secures identity validation and tokens, enforcing role-based permissions.
Paystack Integration	Automates verification of digital payments from buyers.
Arkesel Gateway	Powers transactional, one-way SMS alerts for network-independent updates.
Cloudflare R2	Stores private produce photos and receipts, accessed securely via short-lived signed URLs.

Engineering for Real-World Conditions

- **Low bandwidth:** The public site and farmer portal use server-side rendering and minimal dependencies to load fast on poor connections.
- **Privacy boundaries:** Buyers browse aggregated inventory without access to farmer identities, enforced in packages/permissions.
- **Immutable audit logs:** All changes to quantities, grades, or pricing (spoilage, shrinkage, disputes) are logged for accountability.
- **Financial separation:** Order payments, storage fees, and farmer payouts run in separate state machines, letting admins audit and authorize payouts independently of payment gateways.